Part B Math:

Planer Reconstruction from four views.

Image point in homogeneous coordinates:

A 2D image point in ~~homogeneous co-ordinates~~ written as:

$$x = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

and its co-responding image point in another image is:

$$x' = \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

for a planner object, corresponding points between two views satisfy,

$$x' \sim Hx$$

Where H is a 3x3 homography matrix:

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$

The symbol $\sim$ means equality up to a scaler factor

So,

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \sim \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Multiplying gives,

$$x' = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$$

$$y' = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$$

These equations map a point from one image to the matching point in another image.

A homography has 9 entries, but it is defined only up to scale, so it has 8 independent unknowns. Each point correspondence gives 2 equations.

So, at least 4 corresponding points are needed.

4 points × 2 equations per point = 8 equations

<u>Linear equation setup:</u>

let, $x = (x, y, 1)^T$, $x' = (u, v, 1)^T$

Then form, $x' \sim Hx$

we get two equations,

$$u = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$$

$$v = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$$

Rearranging:

$$h_{11}x + h_{12}y + h_{13} - u h_{31}x - u h_{32}y - u h_{33} = 0$$

$$h_{21}x + h_{22}y + h_{23} - v h_{31}x - v h_{32}y - v h_{33} = 0$$

this becomes,

$$\begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -ux & -uy & -u \end{bmatrix} h = 0$$

$$\begin{bmatrix} 0 & 0 & 0 & x & y & 1 & -vx & -vy & -v \end{bmatrix} h = 0$$

where,

$$h = \begin{bmatrix} h_{11} & h_{12} & h_{13} & h_{21} & h_{22} & h_{23} & h_{31} & h_{32} & h_{33} \end{bmatrix}^T$$

Seeking all correspondence gives

$$Ah = 0$$

<u>Boundary Reconstruction</u>

Suppose the four boundary corners of the planer object in the reference image are:

$P_1 \quad P_2 \quad P_3 \quad P_4$

Then their projected position in another image are:

$$p_i' \sim H p_i, \quad i = 1, 2, 3, 4$$

Connecting these transformed corner points give the estimated boundary of the object in the other view.

Wraping to a common reference plane:

If H maps the reference image to another image, then the inverse homography maps the other image back to the reference plan:

$$x \sim H^{-1} x'$$

this is used for image wrapping and planar reconstruction.

So, the reconstruction is done by mapping all views back to one common plan.

Scanned with CamScanner